

AI-Powered Real-Time Safety Detection System

Figure 1: System Architecture

Illustrates the overall system architecture of the AI-powered real-time safety detection system. The system consists of a high-resolution camera module that continuously captures live video streams of the monitored environment. These video streams are fed into a processing unit equipped with multiple deep learning models that perform concurrent analysis: gender classification using a MobileNet-based classifier, distress hand signal detection leveraging a YOLOv8 object detection model trained on custom gesture datasets, and violence recognition through a convolutional neural network designed to identify aggressive or harmful behaviors. Upon detection of any emergency or suspicious activity, the alert notification system generates real-time warnings that are communicated via IoT platforms or integrated APIs to security personnel, mobile devices, or control centers for immediate response. This integrated architecture ensures continuous monitoring, rapid detection, and timely intervention to enhance user safety.

Figure 2: Block Diagram of Core Components

Illustrates a detailed block diagram of the core components of the AI-powered real-time safety detection system. The process begins with the Camera Module, which continuously captures live video feeds from the monitored area. These raw video frames are sent to the Preprocessing Unit, where essential steps such as resizing, normalization, and noise reduction are applied to prepare the data for efficient analysis by the AI models.

The preprocessed video data is then simultaneously fed into multiple Deep Learning Modules operating in parallel:

- YOLOv8 Module performs rapid object detection and gesture recognition.
- MobileNet Classification Module predicts the gender of each detected individual.
- CNN-based Violence Detection Module analyzes sequences of video frames to detect violent behaviors.

Outputs from these AI modules are passed to the Decision-Making Logic Unit, which integrates the predictions, applies confidence thresholds, and determines whether an alert condition is met. If an alert is triggered, the Output Alert Mechanism activates, sending immediate notifications via IoT communication channels such as mobile apps, SMS, or dashboards. This provides a robust and scalable system for enhancing safety in monitored environments.

AI-Powered Real-Time Safety Detection System

Figure 3: Flowchart Description

1. Video Capture: Camera captures live video feed.
2. Frame Preprocessing: Resize, normalize, enhance video frames.
3. Person and Gesture Detection (YOLOv8): Detect persons and emergency gestures.
4. Gender Classification (MobileNet): Classify detected persons as male or female.
5. Violence Detection (CNN): Identify aggressive or harmful behaviors.
6. Decision Logic: Combine outputs and apply rules to detect emergencies.
7. Alert Generation: Trigger alerts if threat is confirmed.
8. Notification: Send alerts via IoT, API, or mobile devices.
9. User Interface: Display alerts and visual feedback to security personnel.
10. System Feedback: Collect feedback for model improvement.